

GPT-2 Small: corrected final Linear vs ResNet ranking

Layout-safe rebuild: no overflow diagrams; all six Method-2 top-10s; corrected gated + Matryoshka rows.

Data universe

Completed rows: 200
Ranking eligible: 164
Dead-filter omitted: 36

Default #1

ResNet/BatchTopK 8k k=75
EV=0.7768; L0=75.0; H2=53.3

Matched pairs

100 exact family×width×sparsity pairs
Beat=61; Loss=16; Trade=23; Tie=0

Corrected versions used

Gated = shared/tied ResNet-gated (`nonlinear\_resnet\_tied\_gated`).
Matryoshka = corrected multi-ResNet TopK
(`matryoshka\_multi\_resnet\_topk`).

Arch	Fam	Elig	Omit
ResNet	Gated	2	18
ResNet	Std	2	18

Omitted rows are completed runs failing dead_features/d_sae <= 10%; they are omitted from rankings, not missing.

Method-2 six-variant correlation table

Old value-normalized H2 variants; corr = rank correlation vs default global rank-H2.

Variant	corr	top-1	EV	L0	Res@10	BTK@10
raw min/max	0.494	Lin/TopK/8k/k=500	0.9858	500.0	0	4
raw 1-99 pct	0.533	Lin/BTK/8k/k=300	0.9357	300.0	0	4
raw 5-95 pct	0.649	Lin/BTK/8k/k=300	0.9357	300.0	4	4
raw mean±1SD	0.763	Res/BTK/8k/k=200	0.8899	200.0	4	5
log L0 1-99 pct	0.986	Res/BTK/8k/k=100	0.8099	100.0	8	6
log L0 5-95 pct	0.991	Res/BTK/8k/k=100	0.8099	100.0	9	5

Interpretation

High corr means the value-normalized variant preserves the default global EV/L0 H2 ordering. Low corr means the variant changes the tradeoff geometry substantially, so its top-10 deserves a separate slide.

Default top 10: global EV/L0 percentile H2

Rows split 1-5 and 6-10 to prevent table overflow.

ranks 1-5

#	A	Fam	W	Spars	EV	L0	H2
1	Res	BTK	8k	k75	0.7768	75.0	53.3
2	Res	BTK	8k	k100	0.8099	100.0	53.2
3	Res	BTK	8k	k150	0.8592	150.0	53.0
4	Res	TopK	8k	k100	0.7958	100.0	52.6
5	Res	TopK	8k	k150	0.8488	150.0	52.4

ranks 6-10

#	A	Fam	W	Spars	EV	L0	H2
6	Res	Mat	8k	k100	0.7948	100.0	52.3
7	Lin	BTK	8k	k150	0.8431	150.0	52.2
8	Res	BTK	4k	k100	0.7933	100.0	52.0
9	Res	Mat	8k	k150	0.8426	150.0	52.0
10	Res	Mat	8k	k75	0.7645	75.0	51.7

Formula

EV percentile: higher is better. Sparse percentile: lower L0 is better. H2 = harmonic mean of the two percentiles.

Method-2 top 10: raw min/max

Value-normalized EV/L0 H2. Rank correlation vs default global rank-H2 = 0.494.

ranks 1-5

#	A	Fam	W	Spars	EV	L0	H2
1	Lin	TopK	8k	k500	0.9858	500.0	89.7
2	Lin	BTK	8k	k500	0.9855	500.0	89.7
3	Lin	BTK	4k	k500	0.9814	500.0	89.4
4	Lin	TopK	4k	k500	0.9811	500.0	89.4
5	Lin	BTK	8k	k300	0.9357	300.0	89.3

ranks 6-10

#	A	Fam	W	Spars	EV	L0	H2
6	Lin	Mat	8k	k300	0.9331	300.0	89.1
7	Lin	BTK	4k	k300	0.9325	300.0	89.0
8	Lin	Mat	8k	k500	0.9761	500.0	89.0
9	Lin	TopK	8k	k300	0.9309	300.0	88.9
10	Lin	TopK	4k	k300	0.9292	300.0	88.7

Method-2 top 10: raw 1-99 pct

Value-normalized EV/L0 H2. Rank correlation vs default global rank-H2 = 0.533.

ranks 1-5

#	A	Fam	W	Spars	EV	L0	H2
1	Lin	BTK	8k	k300	0.9357	300.0	87.7
2	Lin	Mat	8k	k300	0.9331	300.0	87.4
3	Lin	BTK	4k	k300	0.9325	300.0	87.3
4	Lin	TopK	8k	k500	0.9858	500.0	87.3
5	Lin	BTK	8k	k500	0.9855	500.0	87.3

ranks 6-10

#	A	Fam	W	Spars	EV	L0	H2
6	Lin	TopK	8k	k300	0.9309	300.0	87.2
7	Lin	TopK	4k	k300	0.9292	300.0	87.0
8	Lin	BTK	4k	k500	0.9814	500.0	86.9
9	Lin	TopK	4k	k500	0.9811	500.0	86.9
10	Lin	Mat	8k	k500	0.9761	500.0	86.5

Method-2 top 10: raw 5-95 pct

Value-normalized EV/L0 H2. Rank correlation vs default global rank-H2 = 0.649.

ranks 1-5

#	A	Fam	W	Spars	EV	L0	H2
1	Lin	BTK	8k	k300	0.9357	300.0	84.9
2	Lin	Mat	8k	k300	0.9331	300.0	84.6
3	Lin	BTK	4k	k300	0.9325	300.0	84.5
4	Lin	TopK	8k	k300	0.9309	300.0	84.4
5	Lin	TopK	4k	k300	0.9292	300.0	84.2

ranks 6-10

#	A	Fam	W	Spars	EV	L0	H2
6	Lin	Mat	4k	k300	0.9239	300.0	83.7
7	Res	TopK	8k	k300	0.9196	300.0	83.3
8	Res	Mat	8k	k300	0.9196	300.0	83.3
9	Res	BTK	8k	k200	0.8899	200.0	83.2
10	Res	BTK	8k	k300	0.9147	300.0	82.8

Method-2 top 10: raw mean $\pm$ 1SD

Value-normalized EV/L0 H2. Rank correlation vs default global rank-H2 = 0.763.

ranks 1-5

#	A	Fam	W	Spars	EV	L0	H2
1	Res	BTK	8k	k200	0.8899	200.0	81.8
2	Lin	BTK	8k	k200	0.8849	200.0	81.1
3	Res	TopK	8k	k200	0.8801	200.0	80.4
4	Res	BTK	8k	k150	0.8592	150.0	80.1
5	Res	Mat	8k	k200	0.8773	200.0	80.0

ranks 6-10

#	A	Fam	W	Spars	EV	L0	H2
6	Lin	TopK	8k	k200	0.8769	200.0	79.9
7	Lin	Mat	8k	k200	0.8762	200.0	79.8
8	Lin	BTK	4k	k200	0.8743	200.0	79.5
9	Lin	TopK	4k	k200	0.8704	200.0	78.9
10	Lin	BTK	8k	k300	0.9357	300.0	78.6

Method-2 top 10: log L0 1-99 pct

Value-normalized EV/L0 H2. Rank correlation vs default global rank-H2 = 0.986.

ranks 1-5

#	A	Fam	W	Spars	EV	L0	H2
1	Res	BTK	8k	k100	0.8099	100.0	60.4
2	Res	BTK	8k	k75	0.7768	75.0	59.8
3	Res	BTK	8k	k150	0.8592	150.0	59.6
4	Res	TopK	8k	k100	0.7958	100.0	59.2
5	Res	Mat	8k	k100	0.7948	100.0	59.1

ranks 6-10

#	A	Fam	W	Spars	EV	L0	H2
6	Res	BTK	4k	k100	0.7933	100.0	58.9
7	Res	TopK	8k	k150	0.8488	150.0	58.9
8	Lin	BTK	8k	k100	0.7929	100.0	58.9
9	Lin	BTK	8k	k150	0.8431	150.0	58.5
10	Res	Mat	8k	k150	0.8426	150.0	58.5

Method-2 top 10: log L0 5-95 pct

Value-normalized EV/L0 H2. Rank correlation vs default global rank-H2 = 0.991.

ranks 1-5

#	A	Fam	W	Spars	EV	L0	H2
1	Res	BTK	8k	k100	0.8099	100.0	57.9
2	Res	BTK	8k	k75	0.7768	75.0	57.4
3	Res	TopK	8k	k100	0.7958	100.0	56.6
4	Res	BTK	8k	k150	0.8592	150.0	56.5
5	Res	Mat	8k	k100	0.7948	100.0	56.5

ranks 6-10

#	A	Fam	W	Spars	EV	L0	H2
6	Res	BTK	4k	k100	0.7933	100.0	56.3
7	Lin	BTK	8k	k100	0.7929	100.0	56.3
8	Res	Mat	8k	k75	0.7645	75.0	55.9
9	Res	TopK	8k	k150	0.8488	150.0	55.8
10	Res	TopK	4k	k100	0.7881	100.0	55.8

Matched ResNet-vs-Linear beat/loss by architecture

Each cell is 10 exact matched sparsity settings: same family × same width × same L1 or k.

Fam	W	Beat	Loss	Trade	Tie	mean Δ EV	mean Δ L0
Std	4k	4	0	6	0	-0.1156	+257.1
Std	8k	5	0	5	0	-0.1423	+218.9
Gated	4k	5	0	5	0	-0.0254	+178.6
Gated	8k	2	1	7	0	-0.0267	+176.4
TopK	4k	7	3	0	0	+0.0046	+0.0
TopK	8k	8	2	0	0	+0.0057	+0.0
Mat	4k	7	3	0	0	+0.0079	+0.0
Mat	8k	8	2	0	0	+0.0073	+0.0
BTK	4k	7	3	0	0	+0.0041	+0.0
BTK	8k	8	2	0	0	+0.0069	+0.0

Definitions

Beat = ResNet EV non-worse and L0 non-worse, with at least one strict improvement. Loss is the reverse. Trade means EV and L0 move in opposite directions.

GPT-2 Small: concentration overview for available k-family rows

Lower top-10/top-50 dominance concentration is better: less activation mass is carried by a few broad/high-dominance features.

Fam	W	EV B/L/T	C10 B/L	mean Δ C10	mean Δ C50	Lin C10	Res C10
TopK	4k	7/3/0	5/5	-0.026	-0.039	0.14	0.17
TopK	8k	8/2/0	7/3	-0.006	-0.007	0.15	0.15
Mat	4k	7/3/0	7/3	+0.025	+0.032	0.19	0.17
Mat	8k	8/2/0	9/1	+0.039	+0.049	0.19	0.15
BTK	4k	7/3/0	6/4	-0.005	-0.019	0.20	0.20
BTK	8k	8/2/0	8/2	+0.006	+0.007	0.19	0.18

How to read

C10/C50 are fractions of total activation magnitude explained by the top 10/50 dominance features. A green COut row in the following slides means ResNet is less concentrated than the matched Linear row.

GPT-2 Small: Standard detailed ResNet-vs-Linear rows

All 20 matched rows included, including dead-neuron-bad rows. Within each width: sorted by Linear L0 low→high.

Legend: B=green ResNet beats; L=red loss; T:EV+=orange EV higher/L0 worse; T:L0+=purple L0 lower/EV worse; D%*=dead_frac>10%; ΔL0=Linear L0–ResNet L0.

4k

HP	LinEV	LinL0	LinD%	ResEV	ResL0	ResD%	ΔEV	ΔL0	Out
L5	0.434	20	0%	0.120	2	99%*	-0.314	+19	T:L0+
L4	0.495	37	0%	0.202	9	95%*	-0.294	+28	T:L0+
L3	0.564	72	0%	0.370	39	87%*	-0.194	+34	T:L0+
L2	0.672	171	0%	0.554	108	80%*	-0.118	+63	T:L0+
L1.5	0.742	274	0%	0.628	160	80%*	-0.114	+114	T:L0+
L1	0.829	493	0%	0.695	220	84%*	-0.134	+273	T:L0+
L0.7	0.895	762	0%	0.895	462	61%*	+0.000	+300	B
L0.4	0.957	1172	0%	0.965	766	35%*	+0.008	+406	B
L0.2	0.988	1521	0%	0.990	963	27%*	+0.002	+558	B
L0.1	0.997	1714	0%	0.997	937	41%*	+0.001	+777	B

8k

HP	LinEV	LinL0	LinD%	ResEV	ResL0	ResD%	ΔEV	ΔL0	Out
L5	0.476	22	0%	0.110	1	100%*	-0.365	+20	T:L0+
L4	0.542	40	0%	0.165	6	99%*	-0.377	+34	T:L0+
L3	0.611	81	0%	0.439	52	89%*	-0.171	+29	T:L0+
L2	0.705	193	0%	0.443	74	95%*	-0.262	+119	T:L0+
L1.5	0.767	321	0%	0.486	97	96%*	-0.282	+223	T:L0+
L1	0.842	605	0%	0.844	392	75%*	+0.001	+213	B
L0.7	0.894	970	0%	0.913	784	31%*	+0.020	+186	B
L0.4	0.952	1655	0%	0.962	1388	9%	+0.010	+267	B
L0.2	0.985	2426	0%	0.987	2034	5%	+0.002	+392	B
L0.1	0.996	2906	0%	0.996	2200	13%*	+0.000	+706	B

GPT-2 Small: Gated detailed ResNet-vs-Linear rows

All 20 matched rows included, including dead-neuron-bad rows. Within each width: sorted by Linear L0 low→high.

Legend: B=green ResNet beats; L=red loss; T:EV+=orange EV higher/L0 worse; T:L0+=purple L0 lower/EV worse; D%*=dead_frac>10%; ΔL0=Linear L0–ResNet L0.

4k

HP	LinEV	LinL0	LinD%	ResEV	ResL0	ResD%	ΔEV	ΔL0	Out
L5	0.558	19	0%	0.464	16	70%*	-0.094	+3	T:L0+
L4	0.615	33	0%	0.556	32	69%*	-0.059	+1	T:L0+
L3	0.699	68	0%	0.636	62	73%*	-0.063	+5	T:L0+
L2	0.798	159	0%	0.683	107	83%*	-0.114	+52	T:L0+
L1.5	0.859	261	0%	0.897	263	47%*	+0.038	-2	T:EV+
L1	0.932	477	0%	0.957	456	30%*	+0.025	+20	B
L0.7	0.969	755	0%	0.979	622	27%*	+0.010	+133	B
L0.4	0.992	1130	0%	0.995	822	27%*	+0.002	+307	B
L0.2	0.999	1360	0%	0.999	846	36%*	+0.001	+514	B
L0.1	1.000	1565	0%	1.000	811	52%*	+0.000	+754	B

8k

HP	LinEV	LinL0	LinD%	ResEV	ResL0	ResD%	ΔEV	ΔL0	Out
L5	0.596	19	0%	0.519	19	74%*	-0.077	+0	T:L0+
L4	0.653	35	0%	0.604	37	73%*	-0.049	-2	L
L3	0.725	71	0%	0.491	40	95%*	-0.234	+30	T:L0+
L2	0.813	177	0%	0.843	181	60%*	+0.029	-4	T:EV+
L1.5	0.867	306	0%	0.896	317	34%*	+0.029	-12	T:EV+
L1	0.922	576	0%	0.943	593	15%*	+0.021	-16	T:EV+
L0.7	0.957	896	0%	0.970	918	8%	+0.012	-22	T:EV+
L0.4	0.987	1554	0%	0.989	1304	5%	+0.002	+250	B
L0.2	0.998	2135	0%	0.997	1679	14%*	-0.001	+456	T:L0+
L0.1	1.000	2304	0%	1.000	1220	30%*	+0.000	+1084	B

GPT-2 Small: TopK detailed ResNet-vs-Linear rows

All 20 matched rows included, including dead-neuron-bad rows. Within each width: sorted by Linear L0 low→high.

Legend: B=green ResNet beats; L=red loss; T:EV+=orange EV higher/L0 worse; T:L0+=purple L0 lower/EV worse; D%*=dead_frac>10%; ΔL0=Linear L0–ResNet L0.

4k

HP	LinEV	LinL0	LinD%	ResEV	ResL0	ResD%	ΔEV	ΔL0	Out
k10	0.502	10	0%	0.503	10	0%	+0.001	-0	B
k20	0.567	20	0%	0.586	20	0%	+0.018	-0	B
k30	0.615	30	0%	0.633	30	0%	+0.019	-0	B
k50	0.676	50	0%	0.695	50	0%	+0.019	-0	B
k75	0.725	75	0%	0.747	75	0%	+0.021	-0	B
k100	0.769	100	0%	0.788	100	0%	+0.019	-0	B
k150	0.826	150	0%	0.833	150	0%	+0.007	-0	B
k200	0.870	200	0%	0.861	200	1%	-0.009	-0	L
k300	0.929	300	0%	0.904	300	3%	-0.025	-0	L
k500	0.981	500	1%	0.958	500	3%	-0.023	-0	L

8k

HP	LinEV	LinL0	LinD%	ResEV	ResL0	ResD%	ΔEV	ΔL0	Out
k10	0.534	10	0%	0.544	10	0%	+0.010	-0	B
k20	0.601	20	0%	0.619	20	0%	+0.017	-0	B
k30	0.650	30	0%	0.661	30	0%	+0.011	-0	B
k50	0.702	50	0%	0.719	50	0%	+0.017	-0	B
k75	0.749	75	0%	0.762	75	1%	+0.013	-0	B
k100	0.783	100	0%	0.796	100	1%	+0.013	-0	B
k150	0.837	150	0%	0.849	150	1%	+0.012	-0	B
k200	0.877	200	0%	0.880	200	2%	+0.003	-0	B
k300	0.931	300	0%	0.920	300	4%	-0.011	-0	L
k500	0.986	500	5%	0.957	500	1%	-0.029	-0	L

GPT-2 Small: TopK concentration check

Same matched rows, now asking whether ResNet puts less activation mass in top-dominance features. Lower concentration is better.

Legend: EV/L0 keeps the same colors as the main table. COut: B=green ResNet lower top-10 concentration; L=red higher concentration. ΔC columns are Linear–ResNet, so positive is better for ResNet.

4k

HP	LinEV	LinL0	ResEV	ResL0	EV/L0	LinC10	ResC10	ΔC10	COut	LinC50	ResC50	ΔC50
k10	0.502	10	0.503	10	B	0.28	0.28	+0.00	B	0.43	0.43	+0.00
k20	0.567	20	0.586	20	B	0.23	0.23	+0.00	B	0.35	0.36	-0.01
k30	0.615	30	0.633	30	B	0.19	0.19	+0.00	B	0.32	0.33	-0.01
k50	0.676	50	0.695	50	B	0.17	0.16	+0.00	B	0.31	0.28	+0.02
k75	0.725	75	0.747	75	B	0.13	0.16	-0.02	L	0.27	0.27	+0.00
k100	0.769	100	0.788	100	B	0.11	0.13	-0.02	L	0.24	0.25	-0.01
k150	0.826	150	0.833	150	B	0.09	0.18	-0.09	L	0.20	0.30	-0.10
k200	0.870	200	0.861	200	L	0.08	0.16	-0.08	L	0.17	0.31	-0.13
k300	0.929	300	0.904	300	L	0.06	0.12	-0.06	L	0.14	0.32	-0.19
k500	0.981	500	0.958	500	L	0.06	0.06	+0.00	B	0.25	0.22	+0.03

8k

HP	LinEV	LinL0	ResEV	ResL0	EV/L0	LinC10	ResC10	ΔC10	COut	LinC50	ResC50	ΔC50
k10	0.534	10	0.544	10	B	0.28	0.27	+0.01	B	0.41	0.42	-0.01
k20	0.601	20	0.619	20	B	0.23	0.22	+0.00	B	0.35	0.35	-0.00
k30	0.650	30	0.661	30	B	0.20	0.19	+0.00	B	0.31	0.32	-0.01
k50	0.702	50	0.719	50	B	0.17	0.17	+0.00	B	0.28	0.29	-0.01
k75	0.749	75	0.762	75	B	0.15	0.14	+0.01	B	0.29	0.26	+0.02
k100	0.783	100	0.796	100	B	0.13	0.13	+0.01	B	0.27	0.23	+0.04
k150	0.837	150	0.849	150	B	0.10	0.13	-0.02	L	0.22	0.22	+0.01
k200	0.877	200	0.880	200	B	0.08	0.14	-0.05	L	0.19	0.22	-0.03
k300	0.931	300	0.920	300	L	0.07	0.11	-0.04	L	0.15	0.30	-0.15
k500	0.986	500	0.957	500	L	0.08	0.06	+0.03	B	0.28	0.21	+0.08

Concentration = dominance mass: sum of largest per-feature D_i values. This is a diagnostic, not part of EV/L0 ranking.

GPT-2 Small: Matryoshka detailed ResNet-vs-Linear rows

All 20 matched rows included, including dead-neuron-bad rows. Within each width: sorted by Linear L0 low→high.

Legend: B=green ResNet beats; L=red loss; T:EV+=orange EV higher/L0 worse; T:L0+=purple L0 lower/EV worse; D%*=dead_frac>10%; ΔL0=Linear L0–ResNet L0.

4k

HP	LinEV	LinL0	LinD%	ResEV	ResL0	ResD%	ΔEV	ΔL0	Out
k10	0.489	10	0%	0.505	10	0%	+0.017	-0	B
k20	0.567	20	0%	0.575	20	0%	+0.009	-0	B
k30	0.605	30	0%	0.628	30	0%	+0.023	-0	B
k50	0.667	50	0%	0.692	50	0%	+0.025	-0	B
k75	0.716	75	0%	0.741	75	0%	+0.025	+0	B
k100	0.760	100	0%	0.780	100	0%	+0.020	-0	B
k150	0.821	150	0%	0.829	150	0%	+0.009	-0	B
k200	0.867	200	0%	0.855	200	2%	-0.012	-0	L
k300	0.924	300	0%	0.901	300	2%	-0.022	-0	L
k500	0.972	500	0%	0.958	500	3%	-0.014	-0	L

8k

HP	LinEV	LinL0	LinD%	ResEV	ResL0	ResD%	ΔEV	ΔL0	Out
k10	0.537	10	0%	0.544	10	0%	+0.007	-0	B
k20	0.599	20	0%	0.615	20	0%	+0.016	-0	B
k30	0.639	30	0%	0.657	30	0%	+0.018	-0	B
k50	0.693	50	0%	0.718	50	0%	+0.025	-0	B
k75	0.742	75	0%	0.764	75	0%	+0.022	-0	B
k100	0.781	100	0%	0.795	100	1%	+0.014	-0	B
k150	0.837	150	0%	0.843	150	1%	+0.006	-0	B
k200	0.876	200	0%	0.877	200	3%	+0.001	-0	B
k300	0.933	300	0%	0.920	300	3%	-0.014	-0	L
k500	0.976	500	1%	0.954	500	1%	-0.022	-0	L

GPT-2 Small: Matryoshka concentration check

Same matched rows, now asking whether ResNet puts less activation mass in top-dominance features. Lower concentration is better.

Legend: EV/L0 keeps the same colors as the main table. COut: B=green ResNet lower top-10 concentration; L=red higher concentration. ΔC columns are Linear–ResNet, so positive is better for ResNet.

4k

HP	LinEV	LinL0	ResEV	ResL0	EV/L0	LinC10	ResC10	ΔC10	COut	LinC50	ResC50	ΔC50
k10	0.489	10	0.505	10	B	0.34	0.28	+0.06	B	0.51	0.43	+0.08
k20	0.567	20	0.575	20	B	0.30	0.23	+0.06	B	0.46	0.37	+0.08
k30	0.605	30	0.628	30	B	0.27	0.21	+0.07	B	0.43	0.34	+0.09
k50	0.667	50	0.692	50	B	0.22	0.19	+0.03	B	0.36	0.30	+0.06
k75	0.716	75	0.741	75	B	0.19	0.14	+0.05	B	0.32	0.25	+0.07
k100	0.760	100	0.780	100	B	0.16	0.13	+0.03	B	0.29	0.25	+0.05
k150	0.821	150	0.829	150	B	0.13	0.15	-0.01	L	0.25	0.24	+0.00
k200	0.867	200	0.855	200	L	0.12	0.17	-0.05	L	0.22	0.32	-0.10
k300	0.924	300	0.901	300	L	0.15	0.14	+0.01	B	0.30	0.32	-0.02
k500	0.972	500	0.958	500	L	0.06	0.06	-0.00	L	0.24	0.22	+0.02

8k

HP	LinEV	LinL0	ResEV	ResL0	EV/L0	LinC10	ResC10	ΔC10	COut	LinC50	ResC50	ΔC50
k10	0.537	10	0.544	10	B	0.34	0.27	+0.08	B	0.49	0.42	+0.07
k20	0.599	20	0.615	20	B	0.29	0.22	+0.08	B	0.45	0.35	+0.09
k30	0.639	30	0.657	30	B	0.26	0.20	+0.06	B	0.42	0.32	+0.10
k50	0.693	50	0.718	50	B	0.22	0.17	+0.04	B	0.37	0.29	+0.07
k75	0.742	75	0.764	75	B	0.18	0.14	+0.04	B	0.33	0.26	+0.06
k100	0.781	100	0.795	100	B	0.17	0.12	+0.04	B	0.30	0.23	+0.06
k150	0.837	150	0.843	150	B	0.13	0.10	+0.03	B	0.25	0.19	+0.06
k200	0.876	200	0.877	200	B	0.11	0.12	-0.01	L	0.22	0.20	+0.01
k300	0.933	300	0.920	300	L	0.13	0.12	+0.01	B	0.21	0.31	-0.10
k500	0.976	500	0.954	500	L	0.07	0.05	+0.02	B	0.25	0.20	+0.05

Concentration = dominance mass: sum of largest per-feature D_i values. This is a diagnostic, not part of EV/L0 ranking.

GPT-2 Small: BatchTopK detailed ResNet-vs-Linear rows

All 20 matched rows included, including dead-neuron-bad rows. Within each width: sorted by Linear L0 low→high.

Legend: B=green ResNet beats; L=red loss; T:EV+=orange EV higher/L0 worse; T:L0+=purple L0 lower/EV worse; D%*=dead_frac>10%; ΔL0=Linear L0–ResNet L0.

4k									
HP	LinEV	LinL0	LinD%	ResEV	ResL0	ResD%	ΔEV	ΔL0	Out
k10	0.502	10	0%	0.517	10	0%	+0.015	-0	B
k20	0.573	20	0%	0.588	20	0%	+0.015	-0	B
k30	0.613	30	0%	0.638	30	0%	+0.025	-0	B
k50	0.683	50	0%	0.700	50	0%	+0.017	-0	B
k75	0.733	75	0%	0.755	75	0%	+0.022	-0	B
k100	0.773	100	0%	0.793	100	0%	+0.021	-0	B
k150	0.834	150	0%	0.838	150	0%	+0.004	-0	B
k200	0.874	200	0%	0.859	200	0%	-0.015	-0	L
k300	0.932	300	0%	0.891	300	2%	-0.042	-0	L
k500	0.981	500	1%	0.962	500	6%	-0.020	-0	L
8k									
HP	LinEV	LinL0	LinD%	ResEV	ResL0	ResD%	ΔEV	ΔL0	Out
k10	0.551	10	0%	0.556	10	0%	+0.006	-0	B
k20	0.610	20	0%	0.622	20	0%	+0.012	-0	B
k30	0.659	30	0%	0.674	30	0%	+0.015	-0	B
k50	0.709	50	0%	0.730	50	0%	+0.021	-0	B
k75	0.753	75	0%	0.777	75	0%	+0.024	-0	B
k100	0.793	100	0%	0.810	100	0%	+0.017	-0	B
k150	0.843	150	0%	0.859	150	0%	+0.016	-0	B
k200	0.885	200	0%	0.890	200	0%	+0.005	-0	B
k300	0.936	300	0%	0.915	300	2%	-0.021	-0	L
k500	0.986	500	1%	0.961	500	1%	-0.025	-0	L

GPT-2 Small: BatchTopK concentration check

Same matched rows, now asking whether ResNet puts less activation mass in top-dominance features. Lower concentration is better.

Legend: EV/L0 keeps the same colors as the main table. COut: B=green ResNet lower top-10 concentration; L=red higher concentration. ΔC columns are Linear–ResNet, so positive is better for ResNet.

4k

HP	LinEV	LinL0	ResEV	ResL0	EV/L0	LinC10	ResC10	ΔC10	COut	LinC50	ResC50	ΔC50
k10	0.502	10	0.517	10	B	0.37	0.31	+0.05	B	0.51	0.48	+0.03
k20	0.573	20	0.588	20	B	0.29	0.27	+0.02	B	0.45	0.42	+0.03
k30	0.613	30	0.638	30	B	0.26	0.24	+0.03	B	0.42	0.38	+0.04
k50	0.683	50	0.700	50	B	0.22	0.26	-0.03	L	0.39	0.38	+0.00
k75	0.733	75	0.755	75	B	0.19	0.26	-0.07	L	0.34	0.38	-0.04
k100	0.773	100	0.793	100	B	0.17	0.17	-0.01	L	0.30	0.31	-0.00
k150	0.834	150	0.838	150	B	0.13	0.13	+0.00	B	0.26	0.30	-0.04
k200	0.874	200	0.859	200	L	0.12	0.16	-0.04	L	0.22	0.37	-0.14
k300	0.932	300	0.891	300	L	0.14	0.14	+0.00	B	0.22	0.31	-0.09
k500	0.981	500	0.962	500	L	0.08	0.07	+0.00	B	0.26	0.24	+0.02

8k

HP	LinEV	LinL0	ResEV	ResL0	EV/L0	LinC10	ResC10	ΔC10	COut	LinC50	ResC50	ΔC50
k10	0.551	10	0.556	10	B	0.35	0.32	+0.04	B	0.50	0.47	+0.03
k20	0.610	20	0.622	20	B	0.30	0.27	+0.03	B	0.45	0.42	+0.03
k30	0.659	30	0.674	30	B	0.26	0.24	+0.02	B	0.42	0.38	+0.04
k50	0.709	50	0.730	50	B	0.22	0.20	+0.02	B	0.39	0.34	+0.05
k75	0.753	75	0.777	75	B	0.19	0.15	+0.04	B	0.34	0.29	+0.05
k100	0.793	100	0.810	100	B	0.16	0.13	+0.03	B	0.30	0.26	+0.04
k150	0.843	150	0.859	150	B	0.14	0.16	-0.03	L	0.25	0.28	-0.02
k200	0.885	200	0.890	200	B	0.12	0.21	-0.09	L	0.22	0.36	-0.14
k300	0.936	300	0.915	300	L	0.10	0.10	+0.00	B	0.19	0.28	-0.09
k500	0.986	500	0.961	500	L	0.07	0.07	+0.00	B	0.28	0.21	+0.07

Concentration = dominance mass: sum of largest per-feature D_i values. This is a diagnostic, not part of EV/L0 ranking.

What happened to rows omitted from rankings?

They completed, but failed the dead-feature eligibility gate.

Arch	Fam	W	Elig	Omit	med dead	max dead
ResNet	Gated	4k	0	10	49.5%	83.1%
ResNet	Gated	8k	2	8	31.9%	94.7%
ResNet	Std	4k	0	10	79.9%	99.4%
ResNet	Std	8k	2	8	82.0%	99.6%

Rule

Ranking slides use only rows with `dead_features/d_sae <= 10%`. Beat/loss tables still use all 100 exact matched pairs unless otherwise stated.

This explains the 160/200 OPT and 164/200 GPT eligible counts.

Metric explainer: cosine contribution

How much one pair contributes to a weighted semantic-coherence cosine numerator.

Objects

Example i has embedding e_i and activation a_i for a feature. Example j has embedding e_j and activation a_j . Cosine similarity is $\cos(e_i, e_j)$.

Pair contribution

$$\text{contribution}_{ij} = a_i \times a_j \times \cos(e_i, e_j)$$

Large positive if both examples strongly activate the feature and their text/image embeddings point in similar directions.

How to read it

It is not a final normalized score by itself. It is a numerator term: a highly active, semantically similar pair pushes the weighted-cosine score up; an active but semantically unrelated/opposite pair contributes little or negative.

Tiny example

If $a_i=3$, $a_j=2$, $\cosine=0.8$, $contribution=4.8$. If $\cosine=0.1$, $contribution=0.6$. Same activations, much weaker semantic coherence.

Metric explainer: weighted cosine

Activation-weighted average semantic similarity among examples for one feature.

Formula

$$\text{weighted cosine}(\text{feature } k) = \frac{\sum_{i < j} a_{i,k} a_{j,k} \cos(e_i, e_j)}{\sum_{i < j} a_{i,k} a_{j,k}}$$

Plain English

Look at examples where feature k activates. Pairs where k fires strongly on both examples matter most. Average their embedding cosine similarities.

What high means

High: the feature fires on semantically similar contexts/images. Low: it fires on a mixed bag. Negative: strong activations are on semantically opposed embedding directions.

Difference from unweighted cosine

Unweighted top-K cosine treats selected examples equally. Weighted cosine lets stronger activations dominate, so it asks: when this feature really fires, are those examples semantically coherent?

Metric explainer: Jaccard similarity

Overlap of two features' active-example sets, ignoring activation magnitude.

Formula

$\text{Jaccard}(A, B) = |A \cap B| / |A \cup B|$, where A and B are the examples/tokens where each feature is active.

Plain English

It asks: do these two features turn on for the same examples? 1.0 = always same active set. 0.0 = no shared active examples.

What it is good for

Redundancy / neighbor finding. If two features have high Jaccard, they may cover the same data even if decoder directions or explanations differ.

Difference from weighted cosine

Jaccard is set overlap of firing events. Weighted cosine is semantic similarity among examples, weighted by activation strength. High Jaccard does not require semantically coherent examples; high weighted cosine does.

Readout caveat

This deck is final for Linear 100 vs corrected/final ResNet 100 only.

Included

Linear no-Aux core5: 100 rows. Corrected/final ResNet no-Aux: Standard, tied-Gated, TopK, corrected Matryoshka, BatchTopK = 100 rows.

Not included yet

Pure nonlinear/no-skip and parameter-matched wider-linear controls are separate runners and should not be folded into these claims yet.

Layout note

This safe rebuild intentionally avoids oversized plots; tables are split and abbreviated so text stays inside slide bounds.